

Yuhao Gu · 顾宇浩

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Education

Sun Yat-sen University · Guangzhou, China

2022/9 - 2027/6 (expected)

Computer Science Ph.D.

Supervisor: [Nong Xiao](#), Co-Advisor: [Xianwei Zhang](#)

Soochow University · Suzhou, China

2018/9 – 2022/6

Computer Science B.S.

GPA Rank 1/203, Outstanding Graduate

Research

coMulator: Coordinate Cross-architecture Compilation and Emulation to Accelerate Dynamic Binary Translation

[Under Review](#)

[MICRO'26 (CCF-A)]

Yuhao Gu, Zhongchun Zheng, Nong Xiao, Xianwei Zhang, Yutong Lu

Developed a compiler-emulator collaborative hybrid execution framework to exploit partial portability in source-code. Built on LLVM/QEMU, it is architecture-neutral and supports x86-64, AArch64, and RISC-V, delivering $7.90\times$ speedup over vanilla emulation and 76% native performance.

Patent **ZL202610398741.2** “A Method for Accelerating Simulated Execution of Programs Across Instruction Set Architectures.”

PROVIS: Production Runtime Observability Infrastructure for Engineering Agents

[Under Review](#)

[ATC'26 (CCF-A)]

Yipeng Ouyang, Xin Huang, Bingjie Liu, Zhongchun Zheng, Yuhao Gu, Xianwei Zhang

Built LLM agent evaluation framework on YatCC course – uses multi-dimensional runtime monitoring to expose systematic degradation of software agents in long-horizon workflows.

coMtainer: Compilation-assisted HPC Container Images with Enhanced Adaptability

<https://doi.org/10.1145/3712285.3759790>

[SC '25 (CCF A)]

Yuhao Gu, Haoquan Chen, Xianjie Chen, Jiangsu Du, Zhiguang Chen, Nong Xiao, Xianwei Zhang, Yutong Lu

Resolved container adaptability in HPC – embeds build-time info to enable system-specific toolchains & optimizations; boosts performance by 66.5%–96.3% over generic images on real workloads.

ORFA: Exploring WebAssembly as a Turing Complete Query Language for Web APIs

<https://doi.org/10.1145/3696410.3714826>

[WWW'25 (CCF A)]

Yuhao Gu, Chunyu Chen, Jiangsu Du, Xiaoxi Zhang, Xianwei Zhang

First to use WebAssembly as a query language – leverages its Turing completeness for arbitrary logic in one request, solving the over/under-fetching problem; cuts latency by 18.4% and network traffic by 24.5% vs. GraphQL.

Patent **ZL202510077351.0** “A Programmable Data Exchange System”

Experiences

YatCC: An integrated online platform that delivers Computing-as-a-Service. 2025–NOW

Aggregated computing resource providers such as the Starlight of NSCC-GZ, and vertically integrated user operating environments and experiences using container technology to provide convenient and easy-to-use computing access and application services. This has supported the course experiment teaching of computer systems on campus, and the related work received the Grand Prize in the CCEC 2025 Teaching Case Competition. Related reports: <https://mp.weixin.qq.com/s/UiXcaOTAOur8YTmyMWJ8Q>, https://mp.weixin.qq.com/s/JgbO9i6qM3EPF06_v9tsnw.

Chief Engineer, being fully responsible for key aspects including the project's macro architecture design, technical solution finalization, core code implementation, and development process management.

Infrastructure services and support for the arcSYSu research group. 2025–NOW

Built the group's Kubernetes cluster and deployed internal services including Harbor, GitLab, Nextcloud, etc., as well as the research group's official website. Website: <https://arcsysu.tech>

Primary administrator, designed and set up a full-stack service system architecture from hardware to software, and has been long-term responsible for service operation and maintenance.

SYsU-lang2: Compiler Construction Experiment of SYSU 2023–2024

Designed a compiler frontend/middle/backend construction experiment for a subset of the C language, based on modern infrastructure and toolchains such as LLVM. This work has now been integrated into the YatCC project and received the First Prize in the CCEC 2024 Teaching Case Competition. Project homepage: <https://github.com/arcsysu/SYsU-lang>.

Lead creator, responsible for the overall content design of the current experimental version and the implementation of all core code.

Google Summer of Code 2023 – LLVM Compiler Infrastructure 2023

Add a -show-directory-coverage option to the llvm-cov tool to generate hierarchical coverage reports, better supporting code coverage analysis for large projects. Project homepage: <https://summerofcode.withgoogle.com/archive/2023/projects/ghJRTsUJ>.

Sole contributor, mentored by Petr Hosek and Gulfem Savrun Yeniceri.

Awards

Grand Prize, China Computer Education Conference (CCEC) https://mp.weixin.qq.com/s/l5D3yZIGGC5EkKBH1ju15w	2025
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First Prize, China Computer Education Conference (CCEC)	2024
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CCF Elite Collegiate Award (nationally, 94 in total) https://mp.weixin.qq.com/s/AJKqG2KvLl6n_88_8IzsYw	2021
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Skills

Python, C/C++, LLVM, Kubernetes, OCI Containers, Web,